

Tutorial 6 for Calculus II

Sect. 12.1-12.6

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考点一: Sect. 12.2-12.4

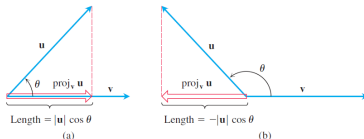
1. **Unit Vector**: If $\mathbf{v} \neq \mathbf{0}$, then

- (1) $\frac{\mathbf{v}}{|\mathbf{v}|}$ is a **unit vector** in the **direction** of $\mathbf{v}$;
- (2) the equation $\mathbf{v} = |\mathbf{v}| \frac{\mathbf{v}}{|\mathbf{v}|}$ expresses $\mathbf{v}$ as its **length** times its **direction**.

2. **Vector Projections**: P724

- (1) The **vector projection** of $\mathbf{u}$ onto $\mathbf{v}$ is the vector $\text{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v}$.
- (2) The **scalar component** of $\mathbf{u}$ in the direction of $\mathbf{v}$ is the scalar

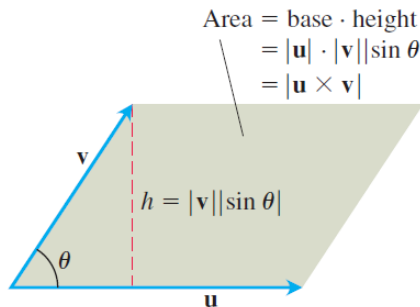
$$|\mathbf{u}| \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} = \mathbf{u} \cdot \frac{\mathbf{v}}{|\mathbf{v}|}.$$



考点一: Sect. 12.2-12.4

3. $|\mathbf{v} \times \mathbf{u}|$ is the Area of a Parallelogram: P729

$$|\mathbf{v} \times \mathbf{u}| = |\mathbf{u}| |\mathbf{v}| |\sin \theta| |\mathbf{n}| = |\mathbf{u}| |\mathbf{v}| |\sin \theta|.$$



考点一: Sect. 12.2-12.4

4. Calculating the Cross Product as a Determinant: P730

If $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ and $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$, then

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$

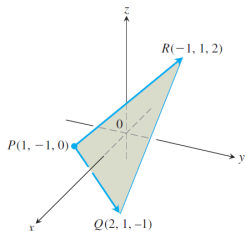


FIGURE 12.31 The vector $\vec{PQ} \times \vec{PR}$ is perpendicular to the plane of triangle PQR (Example 2). The area of triangle PQR is half of $|\vec{PQ} \times \vec{PR}|$ (Example 3).

考点一: Sect. 12.2-12.4

5. Triple Scalar or Box Product: P731

The product $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$ is called the **triple scalar product** of $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$, and

$$|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}| = |\mathbf{u} \times \mathbf{v}| |\mathbf{w}| |\cos \theta|$$

is the **volume** of the parallelepiped (parallelogram-sided box) determined by $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$, $|\mathbf{u} \times \mathbf{v}|$ is the **area** of the base.

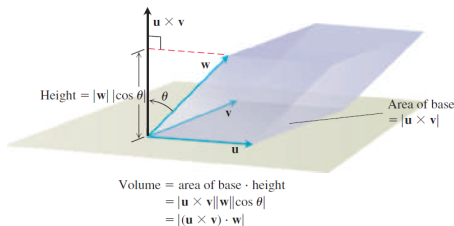


FIGURE 12.34 The number $|(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}|$ is the volume of a parallelepiped.

考点一: Sect. 12.2-12.4

6. Calculating the Triple Scalar Product as a Determinant: P732

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w} = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}.$$

相关考题

- (2019年期中) Let $\vec{a}$ and $\vec{b}$ be two nonzero orthogonal vectors, which of the following must be true?
(A) $|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$; (B) $|\vec{a} - \vec{b}| = |\vec{a}| - |\vec{b}|$
(C) $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$; (D) $\vec{a} + \vec{b} = \vec{a} - \vec{b}$
- Determine whether the following statements are true or false? No justification is necessary.
(1) (2020年期末) If $\vec{u} \neq 0$, and if $\vec{u} \times \vec{v} = \vec{u} \times \vec{w}$ and $\vec{u} \cdot \vec{v} = \vec{u} \cdot \vec{w}$, then $\vec{v} = \vec{w}$.
- (2021年期中) If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors and $\vec{a} + \vec{b} + \vec{c} = 0$, then $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} = \underline{\hspace{2cm}}$.
- (2021年期中) If the vector $\vec{c}$ is perpendicular to $\vec{a} = (1, 2, 1)$ and $\vec{b} = (-1, 1, 1)$ and $\vec{c} \cdot (\vec{i} - 2\vec{j} + \vec{k}) = 16$, then $\vec{c} = \underline{\hspace{2cm}}$.

考点二: Vector Equation for a Line: Sect. 12.5

1. Vector Equation for a Line: P734

(1) A vector equation for the line L through $P_0(x_0, y_0, z_0)$ parallel to $\mathbf{v}$ is

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}, \quad -\infty < t < \infty$$

(2) Parametric Equations for a Line: P734

The standard parametrization of the line through $P_0(x_0, y_0, z_0)$ parallel to $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$ is

$$x = x_0 + tv_1, y = y_0 + tv_2, z = z_0 + tv_3, \quad -\infty < t < \infty$$

(3) The line segment: we first parametrize the line through the points.

Then find the t -values for the endpoints and restrict t to lie in the closed interval bounded by these values.

考点二: Vector Equation for a plane: Sect. 12.5

2. Equation for a Plane: P737

The plane through $P_0(x_0, y_0, z_0)$ normal to $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ has

Vector equation: $\mathbf{n} \cdot \overrightarrow{P_0P} = 0$

Component equation: $A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$

Component equation simplified: $Ax + By + Cz = D$, where

$$D = Ax_0 + By_0 + Cz_0.$$

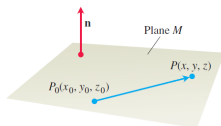


FIGURE 12.39 The standard equation for a plane in space is defined in terms of a vector normal to the plane: A point P lies in the plane through P_0 normal to $\mathbf{n}$ if and only if $\mathbf{n} \cdot \overrightarrow{P_0P} = 0$.

相关考题

1. (2020年期末) The equation of the plane through the line $x = -1 + 2t, y = 3 + t, z = -t$ and parallel to the line $x = -2t, y = t, z = 1 - t$ is _____.
2. (2023年期中) Let $L_1 : x = t, y = 1, z = 1 + t$ and $L_2 : x = 1 + 2s, y = 0, z = 2s$. Is there one plane in which both lines lie ? If so, find the equation of the plane. If not, give your reason.
3. (2022年期末) Find the equation for the plane through the origin parallel to the following lines: $l_1 : x = 1, y = -1 + t, z = 2 + t$, and $l_2 : x = -1 + t, y = -2 + 2t, z = 1 + t$.

相关考题

4. (2021年期末) Find the equation of the plane through point $(1, 0, 1)$, and perpendicular to the plane $x - 2y + 3z + 2 = 0$ and the plane $x + 2y - 3z - 2 = 0$.
5. (2019年期中) Find the equation of the plane through the points $(2, -1, -1)$ and $(1, 0, -1)$ perpendicular to the plane $2x + 3y - 5z + 6 = 0$.

考点三: The Distance: Sect. 12.5

1. The Distance:

(1) Distance from a Point S to a Line Through P Parallel to $\mathbf{v}$: P736

$$d = |\overrightarrow{PS}| \sin \theta = \frac{|\overrightarrow{PS} \times \mathbf{v}|}{|\mathbf{v}|}$$

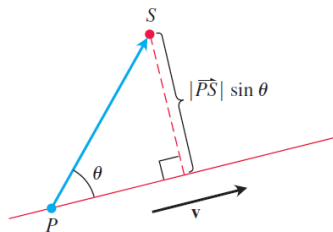


FIGURE 12.38 The distance from S to the line through P parallel to $\mathbf{v}$ is $|\overrightarrow{PS}| \sin \theta$, where θ is the angle between $\overrightarrow{PS}$ and $\mathbf{v}$.

考点三: The Distance: Sect. 12.5

(2) The Distance from a Point to a Plane : P739

$$d = |\overrightarrow{PS}| \cos \theta = \left| \frac{\overrightarrow{PS} \cdot \mathbf{n}}{|\mathbf{n}|} \right|$$

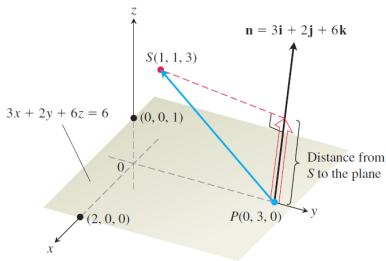


FIGURE 12.41 The distance from S to the plane is the length of the vector projection of $\overrightarrow{PS}$ onto $\mathbf{n}$ (Example 11).

考点三: Angles Between Planes Sect. 12.5

2. Angles Between Planes: P740

The angle between two intersecting planes is defined to be the **acute angle** between their **normal vectors**.

$$\theta = \cos^{-1}\left(\frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{|\mathbf{n}_1||\mathbf{n}_2|}\right), \quad 0 \leq \theta \leq \frac{\pi}{2}.$$

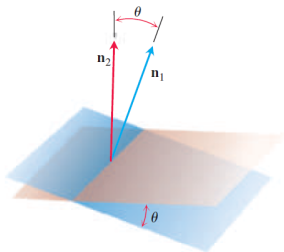


FIGURE 12.42 The angle between two planes is obtained from the angle between their normals.

相关考题

1. (2019年期中) The equations of two lines are

$$l_1 : x = t, y = 2t, z = -t, \text{ and } l_2 : x = 1 - 2t, y = t, z = -1 + t.$$

Then l_1 and l_2 are

- (A) parallel; (B) orthogonal;
(C) intersect with each other; (D) skew.

2. (2023年期中) If P_1 and P_2 are the following planes:

$$P_1 := \{(x, y, z) : x + 2y - z = \sqrt{2}/2\},$$

$P_2 := \{(x, y, z) : x + y = -\sqrt{2}/2\}$, then the angle θ between P_1 and P_2 is:

- (A) $\frac{\pi}{2}$; (B) $\frac{\pi}{3}$; (C) $\frac{\pi}{4}$; (D) $\frac{\pi}{6}$.

3. Determine whether the following statements are true or false? No justification is necessary.
- (1) (2019年期末) The plane $x + y - 2z = 1$ is perpendicular to the plane $x + y + z = 1$.
4. (2023年期中) The distance from the point $S(1, 1, 1)$ to the plane $x + 2y + 3z + 4 = 0$ is ____.

王老师班补充作业5,6

1. Prove all three altitudes of a triangle always intersect at the same point.
9. Suppose the plane M passes through the origin and $(6, -3, 2)$ and is perpendicular to the plane $4x - y + 2z - 8 = 0$. Find M .
14. Find the set of equidistant points to $(3, 1, 2)$, $(4, -2, -2)$ and $(0, 5, 1)$, and describe the geometric of this set.
18. Give line $l : x = 1 + t, y = -1 - t, z = t$ and plane $S : 3x - y + z = 5$:
 - (a) Find the plane passing l and perpendicular to S ;
 - (b) Find the equation of the projection of l to S .
19. Find the projection of the following line $x + 5y + 6z = 3$,
 $x - 2y + 3z = 1$ to the XOY-plane.

花老师班补充作业6

1. Determine whether the following statements are true or false? No justification is necessary.

(1) Equation $r = 2 \sin(\theta)$ ($0 \leq \theta \leq \pi$) in polar form is a circle of radius 1 centered at $(0,1)$.

(3) If $\mathbf{u} \neq 0$, and if $\mathbf{u} \times \mathbf{v} = \mathbf{u} \times \mathbf{w}$ and $\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{w}$, then $\mathbf{v} = \mathbf{w}$.

- 3(3). The equation of the plane through the line

$x = -1 + 2t, y = 3 + t, z = -t$ and parallel to the line

$x = -2t, y = t, z = 1 - t$ is ().

3. Find the equation of the plane through point $(1, 0, 1)$, and perpendicular to the plane $x - 2y + 3z + 2 = 0$ and the plane $x + 2y - 3z - 2 = 0$.

花老师班补充作业6

8. Prove the distance between a point (x_0, y_0, z_0) to the plane $Ax + By + Cz + D = 0$ is given by $d = \frac{|Ax_0 + By_0 + Cz_0 + D|}{\sqrt{A^2 + B^2 + C^2}}$.
17. Let $\overrightarrow{AB} = (-3, 0, 4)$ and $\overrightarrow{AC} = (5, -2, -14)$. Find a unit vector in the direction of the bisector of $\angle BAC$.
18. Give line $l : x = 1 + t, y = -1 - t, z = t$ and plane $S : 3x - y + z = 5$:
- (a) Find the plane passing l and perpendicular to S ;
 - (b) Find the equation of the projection of l to S .